GOVERNMENT PROCUREMENT TENDER SPECIFICATION

Tender Ref: BRO/TUNNEL/PPE-HELMET/2026/12 | Dept: Border Roads Organisation (BRO) - Strategic T

Supply of Industrial Safety Helmets for Tunnel Boring Project Workers

BORDER ROADS ORGANISATION - MINISTRY OF DEFENCE

NIT Reference: BRO/TUNNEL/PPE-HELMET/2026/12
Project: Rohtang Bypass and High-Altitude Tunnel Works

TECHNICAL SPECIFICATIONS - PERSONAL PROTECTIVE EQUIPMENT (PPE):

1. ITEM SPECIFICATION:

Supply of Non-Metallic Industrial Safety Helmets with 4-point harness and sweatband for construction workers, miners, and engineers exposed to falling objects and electrical shock hazards.

2. APPROVED MAKES AND BRANDS:

Helmets shall be procured from reputed safety brands such as Karam or 3M or Karam Safety. Alternate local makes must demonstrate equivalent certification.

3. PERFORMANCE AND TEST REQUIREMENTS:

- Shell Material: High density virgin polymer (HDPE/ABS) resistant to impact, puncture, and UV degradation.
- Shock Absorption: Transmitted force through the helmet shell to the headform shall not exceed 5.0 kN when subjected to 5 kg falling striker drop test from 1.0 meter height.
- Penetration Resistance: Conical striker dropped from 1.0 m shall not make contact with headform.
- Electrical Insulation: Shell shall withstand high-voltage electrical breakdown test of 2000 Volts AC leakage current <= 3 mA.
- Flame Resistance: Shell material shall not continue burning for more than 5 seconds after burner flame removal.
- Chin Strap Retention: Chin strap with minimum 19 mm width with breaking load between 150 N and 250 N.
- Regulatory Mandate: Helmets must bear mandatory ISI Mark under the Personal Protective Equipment Quality Control Order issued by DPIIT.